QUALIFYING EXAMINATION, Part 4

Solutions

Problem 1: Statistical Mechanics I

(a) The single-particle partition function Z_1 is given by

$$Z_1 = \int \frac{d^3x\, d^3p}{h^3}\, e^{-\beta E} = \int \frac{d^3x\, d^3p}{h^3}\, e^{-\beta\left(\frac{p_x^2}{2m}+\frac{p_y^2}{2m}+\frac{p_z^2}{2m}\right)} ,$$

which yields

$$Z_1 = \frac{Ax}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2} .$$

(b) For N non-interacting indistinguishable particles we have

$$Z_N = \frac{Z_1^N}{N!} = \frac{A^N x^N}{N! h^{3N}}\left(\frac{2\pi m}{\beta}\right)^{3N/2} .$$

(c) The partition function Z of the N particles plus lid is given by

$$Z = \frac{1}{h}\int \frac{dx\, dp_{\mathrm{lid}}}{h} Z_N(x) e^{-\beta\left(Fx+\frac{p_{\mathrm{lid}}^2}{2M}\right)} ,$$

where we have integrated over coordinates and momenta of the N particles to obtain $Z_N(x)$.

The integral over the lid momentum p_{lid} gives

$$\int dp_{\mathrm{lid}} e^{-\beta p_{\mathrm{lid}}^2/2M} = \left(\frac{2\pi M}{\beta}\right)^{1/2} ,$$

while the integral over the lid position can be written as

$$\int dx Z_N(x) e^{-\beta F x} = \frac{A^N}{N! h^{3N}}\left(\frac{2\pi m}{\beta}\right)^{3N/2}\int dx x^N e^{-\beta F x} .$$

Substituting $u = \beta F x$ and using the given formula we find

$$\int dx x^N e^{-\beta F x} = \frac{1}{(\beta F)^{N+1}}\int_0^\infty du\, u^N e^{-u} = \frac{N!}{(\beta F)^{N+1}} .$$

The partition function of particles plus lid is then

$$Z = \left(\frac{2\pi M}{\beta}\right)^{1/2}\frac{A^N}{h^{3N+1}}\left(\frac{2\pi m}{\beta}\right)^{3N/2}\frac{1}{(\beta F)^{N+1}} .$$

(d) The mean height is given by

$$\langle x \rangle = -\frac{1}{\beta}\frac{d}{dF}\ln Z = (N+1)\frac{k_B T}{F} \ .$$

(e) The variance in height is given by

$$\langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{\beta^2}\frac{d^2}{dF^2}\ln Z = (N+1)\frac{(k_B T)^2}{F^2} \ .$$

(f) The average spacing between particles is $\left(\frac{V}{N}\right)^{1/3}$. Quantum effects are important when

$$\left(\frac{V}{N}\right)^{1/3} \leq \lambda = \frac{h}{\sqrt{2\pi m k_B T}} \ .$$

We have

$$\frac{V}{N} \sim \frac{\langle x \rangle A}{N} = \frac{N+1}{N}\frac{A k_B T}{F} \approx \frac{A k_B T}{F} \ .$$

Substituting in the inequality above, we find that quantum effects are important for temperatures

$$k_B T \leq \frac{h^{6/5} F^{2/5}}{A^{2/5}(2\pi m)^{3/5}} \ .$$

Problem 2: Statistical Mechanics II

(a) $n_\sigma(\mathbf{k})$ is the grand canonical Bose-Einstein occupation factor for a state of energy $\frac{\hbar^2\mathbf{k}^2}{2m} - \mu_{\rm b}\sigma B$.

(b) $N_\sigma = \Sigma_{\mathbf{k}} n_\sigma(\mathbf{k})$. Taking the continuum limit, we find

$$\begin{aligned} N_\sigma &= \frac{V}{(2\pi)^3}\int \mathrm{d}^3k \frac{1}{\exp\left(\beta(\frac{\hbar^2\mathbf{k}^2}{2m} - \mu_{\rm b}\sigma B - \mu)\right) - 1} \\ &= \frac{V}{2\pi^2}\int \mathrm{d}k \frac{k^2}{\exp\left(\beta(\frac{\hbar^2k^2}{2m} - \mu_{\rm b}\sigma B - \mu)\right) - 1} \\ &= \frac{V}{\lambda^3}F_{3/2}(\zeta e^{\beta\mu_{\rm b}\sigma B}) \ , \end{aligned}$$

The last line is obtained by using the substitution $x^2 = \beta\frac{\hbar^2k^2}{2m}$ in the integral and using the formula for $F_r(z)$ in 'useful formulas' with $r = 3/2$, $z = \zeta e^{\beta\mu_{\rm b}\sigma B}$, and $\zeta = e^{\beta\mu}$ the fugacity. The magnetization is then

$$M = \frac{V\mu_{\rm b}}{\lambda^3}[F_{3/2}(\zeta e^{\beta\mu_{\rm b}B}) - F_{3/2}(\zeta e^{-\beta\mu_{\rm b}B})] \ .$$

(c) The dimensionless small parameter to expand at low B is $\beta\mu_{\rm b}B$. Expanding the above expression to lowest order in $\beta\mu_{\rm b}B \ll 1$ yields (using the corresponding equation in 'useful formulas')

$$M \approx \beta\mu_{\rm b}B\frac{2\mu_{\rm b}V}{\lambda^3}F_{1/2}(\zeta) \ .$$

Differentiating with respect to B, we find

$$\chi = \beta\mu_{\rm b}\frac{2\mu_{\rm b}V}{\lambda^3}F_{1/2}(\zeta) \ .$$

(d) The critical temperature $T_{\rm c}$ for BEC at $B = 0$ is determined by $\rho\lambda_c^3 = 3F_{3/2}(1)$, where ρ is the total density and λ_c is defined for $T = T_c$. The critical temperature $T_{\rm c}$ is found by inverting this relation, i.e., $T_{\rm c} \propto \rho^{2/3}$. Using the given limit $\lim_{\zeta\to 1^-} F_{1/2}(\zeta) = \infty$, we find that $\chi \to \infty$ as $T \to T_{\rm c}^+$, so the susceptibility diverges as T approaches $T_{\rm c}$ from above.

(e) To find μ we note that all $n_\sigma(\mathbf{k})$ must be positive; thus μ has to be bounded from above by the lowest possible single-particle energy, which is $-\mu_{\rm b}B$ for $\mathbf{k} = \mathbf{0}$ and $\sigma = +1$. Thus the chemical potential for $T < T_{\rm c}$ is $\mu = -\mu_{\rm b}B$ and the macroscopic BEC occupation occurs in the lowest energy single-particle lowest state $\mathbf{k} = \mathbf{0}$ and $\sigma = +1$.

(f) Since it is the population $\sigma = +1$ that gets macroscopically occupied, the total magnetization in the limit $B \to 0$ is given by $M = \mu_{\rm b}N_+ = \mu_{\rm b}(N - N_{\rm exc}) = \mu_{\rm b}[N - \frac{3V}{\lambda^3}F_{3/2}(1)]$, where $N_{\rm exc}$ is the population in the excited $\mathbf{k} \neq \mathbf{0}$ states. This result is nonzero for $T < T_c$ and is an example of spontaneous symmetry breaking: the system forms spontaneously a ferromagnet, and any small non-zero magnetic field would result in a non-zero macroscopic magnetic moment.